BEFORE THE
NEW YORK STATE
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.	)	Case Number: 19-G-0309
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service.	)	Case Number: 19-G-0310

DIRECT TESTIMONY TESTIMONY OF BOB WYMAN

August 30, 2019

Introduction

Q. What is your name?

A. I am commonly known as Bob Wyman. My formal name is Robert Mark Wyman.

Q. Mr. Wyman, have you previously testified before the Commission?

A. Yes. I presented testimony during the recent Central Hudson, Orange & Rockland, and Consolidated Edison rate cases. I have also given verbal testimony at a variety of PSC public hearings, have helped others prepare testimony for a variety of PSC proceedings, and am an active party in a variety of PSC matters.

Q. Do you always participate in PSC matters as an individual, representing only yourself?

A. No. In addition to participating in PSC proceedings as an individual, I have, from time to time, assisted others in preparing for these proceedings or have spoken for them. For instance, I have several times assisted NY-GEO[1] in developing testimony and/or evidence. I also represented NY-GEO as a participant in ConEd's "Peak Gas Collaborative" which had been established as a result of the prior ConEd rate case (16-G-0061).[2].

Q. What is your background?

A. I am a New York City-based advocate for renewable energy and Beneficial Electrification with a focus on geothermal heat pumps. I am a member of the New York Geothermal Energy Organization, (NY-GEO)[3], the Renewable Thermal Alliance[4], and the International Ground Source Heat Pump Association (IGSHPA)[5] on whose Advo-

[1] https://ny-geo.org/

[2] 16-G-0061, Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Gas Service.

[3] https://ny-geo.org/

[4] https://cbey.yale.edu/programs-research/renewable-thermal-alliance

[5] https://igshpa.org/

cacy Committee I serve. I am a founder and shareholder, but not an employee, of Dandelion Energy[6], the Google X[7] spin-off company that currently installs geothermal heat pump systems in New York State.

Although most of my career was spent in the computer software business, during which I worked for a variety of companies, including Digital Equipment Corporation, Microsoft, Google, Medio Multimedia, and Pubsub Concepts,[8] I have been advocating on energy related issues for at least 45 years, having first given public testimony in hearings concerning Pres. Nixon's "Project Independence"[9] in 1973.

[6] https://dandelionenergy.com/
[7] https://x.company/
[8] See my LinkedIn profile for more detail: https://www.linkedin.com/in/bobwyman/
[9] https://en.wikipedia.org/wiki/Project_Independence

Purpose of Testimony

Q. What is your purpose in submitting this testimony?

A. I wish to comment on several aspects of the proposals made by National Grid ("The Company") in the current cases and to advocate for several modifications or additions to those proposals. My testimony is intended to ensure a more complete record of the relevant issues and to advocate for policies that will be more fair, more in the public interest, and more likely than those proposed by the Company to conform to existing State and New York City laws and policies.

Some of the issues that I wish to discuss include:

- The Company's proposals concerning
  - Gas expansion
  - Depreciation
  - Excess Deferred Income Tax (EDFIT).
  - Planning a "Process of Managed De-Capitalization"

Gas Expansion and Gas Capital Investment

Q. Should the Company's proposed gas expansion projects be accepted?

A. No. This is undoubtedly the last rate case during which the Company may even hope that it has the opportunity to receive approval for either gas expansion or for any substantial gas capital investment, other that required for safety and reliability. For all the reasons that will prevent the Company, and any other gas utility in New York State, from obtaining approval for gas expansion in future rate cases, the Company's proposals in this rate case should be, at least, severely limited — if not rejected.

Q. Why are future approvals for gas expansion unlikely?

A. A continuously growing public awareness of the full climate, health and economic costs of burning fossil fuels, including gas, has already risen to such a level that public authorities, both at the State and City level, have been compelled to enact laws, policies and regulations that require a dramatic reduction in the future use of these fuels. The goal of these laws is to ensure cleaner, healthier air and water, a reduction in climate impacts, and the adoption of more efficient and economical methods for accomplishing those things currently done with fossil fuels such as gas. The result is an unprecedented regulatory and market environment for the Company.

In the past, the Company has always been able to assume broad government and public support for increasing its customer base, increasing its capacity to serve, and for making long-term gas capital investments. In the past, the Company has also always been able to assume that the assets they invested in would be used at, or close to, full capacity over all or most of their technical lives. However, our new laws and policies now demand that the Company's focus shift from seeking perpetual growth to implementing a process of managed decline. The law, policies and public opinion now require that the Company focus on reducing the number of customers they serve, reducing the quantity of gas they supply, and crafting a process of managed decapitalization in order to avoid the accumulation of stranded assets as we find the useful life of assets ending much earlier than was typical in the past..

Q. Why are you so sure that gas sales must fall in the future?

A. New York State and New York City have both established aggressive non-optional goals limits on carbon emissions. State law requires an 85% reduction in emissions by 2050 and New York City law requires an 80% reduction. New York City's limits are defined in Local Law 97-2019 and are expressed using calendar year 2005 as the base

year. In addition to the 80% reduction by 2050, the City also requires a 30% reduction by 2030.

Although State law requires dramatic emissions reductions by 2050, one might argue that it is possible that emissions reductions might be uneven throughout the state. Thus, KEDNY or KEDLI would be able to reduce emissions by less than 80% if emissions from sources in other parts of the state are reduced by some higher amount. However, KEDNY's emissions are controlled by both State and New York City law thus, reductions elsewhere in the State will have no impact on KEDNY's requirement to comply with City law. Given that only KEDNY and ConEd provide natural gas in New York City, it is useful to look at the impact of New York City law on KEDNY's prospects in the future. In New York City, KEDNY will only be able to reduce its sales reductions if it can shift the burden of reduction to ConEd...

According to the New York City Greenhouse Gas Inventory,[10] in 2005, the City's carbon emissions from all sources was 61,062,452 tCO2e/year. Thus, the City's law requires that carbon emissions in 2030 be reduced to 42,743,716 tCO2e/year and by 2050 to no more than 12,212,490 tCO2e/year. However, **the inventory of NYC's 2017 emissions reports that carbon emissions attributable to natural gas use in buildings[11] were 17,639,988 tCO2e or 144% of the 2050 city-wide, all-sources limit and 41% of the 2030 limit.** (It should be recognized that natural gas use in New York City has increased since 2017. Thus, when the Inventory is updated for 2019, it will show that emissions from natural gas are even greater than 144% of the 2050 limit.) Given that carbon emissions from natural gas already dramatically exceed the City's limit on carbon emissions from all sources, it is clear that emissions from natural gas must

[10] https://on.nyc.gov/ghginventory
[11] "Stationary Source" not including power generation.

be reduced. If we assume that in 2050, natural gas will be the only source of carbon emissions in the City (a fantastic assumption) it would be necessary to reduce natural gas use by 31% in order to avoid exceeding the limits established by City law. Of course, if we assume, as we must, that there will be other sources of carbon emissions, it is clear that natural gas use must be reduced by more than 31% before 2050.

Q. Have carbon emissions from buildings been declining in New York City?

A. Yes, as shown in Figure 1, emissions from buildings for all energy sources other than natural gas have decreased by 21% since 2005. Emissions from oil use are down 53%, emissions from electricity use are down 45% and emissions from steam are down 58%. However, the 35% increase in emissions from natural gas has limited the reduction in this sector's total emissions.

Q. Will further reductions in emissions from other sources or by improving energy use efficiency allow avoidance of reductions in natural gas emissions?

A. No. As stated above, natural gas emissions are already at a level equal to or higher than 144% of the citywide, all-sources limit on carbon emissions for 2050. Thus, even if emissions from all other sources in buildings, as well as all other sources in all sectors (including transportation) were completely eliminated, it would still be necessary to dramatically reduce emissions from natural gas. Similarly, improvements in energy efficiency, such as higher efficiency furnaces, window treatments, or increased insulation would not allow us to avoid dramatic reductions in natural gas use. Such improvements may reduce the amount of emissions from non-natural gas sources and thus might minimize the total reduction required, but emissions from natural gas must still be reduced by at least 30%.

Q. Is it necessary for gas use to be reduced 80% by 2050 in order to meet the requirements of Local Law 97-2019?

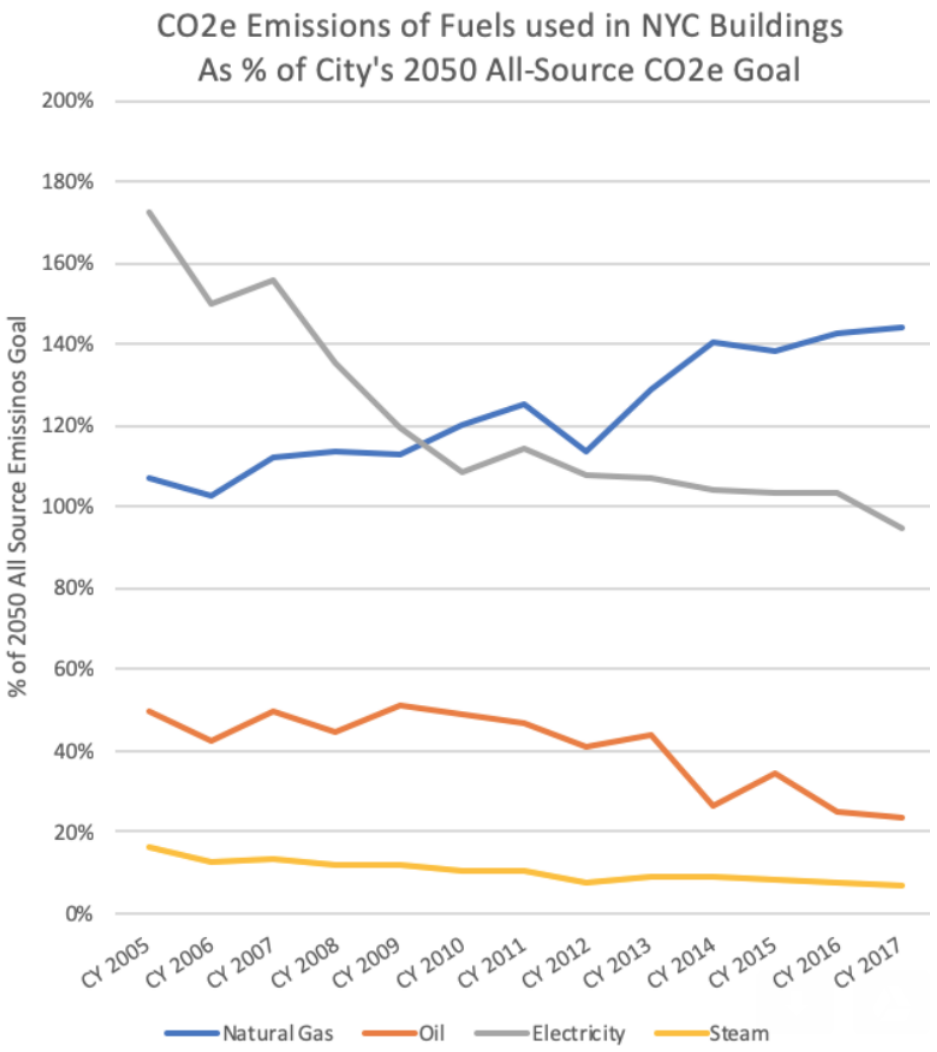


Figure 1: Emissions from Buildings in New York City (Source: NYC Emissions Inventory)

A. No. Although doing so would be desirable, in order for the City's citywide, all-source limits to be met it is only necessary to reduce emissions from natural gas by some quantity larger greater than 30% of the City's natural gas consumption in 2017. How much greater than 30% would be determined by the emissions produced by others sources and by how much natural gas use has risen since 2017. A 30% reduction would be sufficient only if natural gas were to be the only source of carbon emissions in 2050 and if gas expansion had ended in 2017. It is likely that a reduction of between

50% and 80% would be necessary to allow the City's limits to be met.

Q. Given the current law and policies, why is the Company proposing gas capital investment?

A. Because it is all they know to do... The Company has an obligation not only to the public but also to its shareholders. Those shareholders require that the Company be able to show that it has the ability to produce long term profit streams that will benefit its shareholders. However, given the current regulatory requirements, the only means the Company has to produce revenues is to invest in capital assets, from which it earns an allowed return on equity.

Gas utilities are not, in fact, in the "business of selling gas." At least, that isn't how they make money. Gas utilities sell gas to their customers at, or near, that commodity's cost. Additionally, even though they spend a great deal to operate and manage their large and complex transmission and distribution systems, those O&M expenses are paid by ratepayers at cost – without a profit. The only way a gas utility makes money in its regulated business is from their allowed return on the equity they invest in capital assets. Thus, in order to assure shareholders that there will be future profits, the Company must always be able to demonstrate a plan and ability to ensure that shareholder's capital is, and will remain, fully invested in capital assets that deliver a reasonable return on equity. In the past, this has not been a challenge since it has always been possible to obtain approval for additional and ever-growing capital investments. As old assets were retired or depreciated the Company was always able to invest in new gas assets to replace the old. But, times have changed. Continued investment in a large, expensive system whose use is required to decline is simply no longer reasonable.

Given today's laws, policies, etc., it is clear that continued investment in gas assets is not in the public interest, however, such investments are necessary to convince shareholders to maintain their investments in the Company. The conflict between the interests of the Company and those of shareholders appears to be significant. However, we should not be surprised if the Company seeks to prioritize shareholders' interests.

Q. If continued investment is in shareholders' interest, what policies would be most in the public interest?

A. To serve the public interest, and to conform to the law, the Company should:

- Establish a formal, managed moratorium or permanent end to gas expansion
- Establish a year-by-year "glide-path" for reducing gas consumption that conforms to the various legally expressed limits on greenhouse gas emissions in 2030, 2050, etc.
- Create programs, incentives, etc. to ensure that gas consumption stays at or below the levels defined by the glide-path over its full extent.
- Accelerate depreciation and thus revenue-recovery from ratepayers for assets whose useful lives will be shorter than was historically typical.
- The Company should petition the Commission and the Legislature to eliminate the "100 foot rule" which currently forces the Company to provide service on-demand, no matter how much doing so may be against the public interest.

Q. What do you mean by "managed moratorium?"

A. Unlike the current, informal moratorium during which the Company has simply refused to process new applications for service, the Company should do what was done by ConEd in Westchester County and announce a deadline for the acceptance of new applications for gas service. This will allow those who have projects under construc-

tion or in design, but have not yet applied for gas service, to determine whether or not they will be able to receive gas service in the future. The Company should require that applicants demonstrate a capacity to accept service within one year of having that service approved.

Once the initial collection of applications have been accepted, the Company should accept applications only from those customers who can prove that they have special needs or no viable alternative to gas service – given the nature of their usage requirements. Those customers should then only be granted service if it can be shown that sufficient demand has been relieved in order to ensure that the new gas consumption will leave total consumption below that which is required by the "glide-path."

Q. How should be glide-path be defined?

A. The precise shape of the glide-path should be proposed by the Company, in cooperation with parties, after a detailed study of the effectiveness of potential incentives, legal mandates, workforce development, etc. A simple start might be to assume a straight-line decline over the thirty years remaining between now and 2050. Clearly, various parties will advocate for alterations to such a curve, probably to assume greater declines in future years and less decline in near-term years. The actual glide-path should be the result of extensive study and stakeholder discussion.

Q. Why must cost recovery be accelerated?

A. An example might best show the need: The Company currently estimates that services, the connections between customers and mains, will have an average service life of between 55 and 60 years, depending on the type of pipe used. However, it is clear that between now and 2050, only thirty years from now, many customers will terminate gas service and switch to cleaner, cheaper, more sustainable alternatives, such as heat pumps. As those customers switch away from gas, their services will be re-

tired before expected. (Note: "Services" only have one customer. They are not shared assets.) Thus, the actual Average Service Life for these abandoned assets will be less than that which was historically common. In order to ensure that stranded assets (including un-depreciated assets not in use) do not accumulate, it will be necessary to reduce Average Service Life expectations in order to avoid an inter-generational cost-shift to the ever-declining number of future ratepayers.

The amount by which cost recovery is accelerated should be determined upon careful consideration of the glide-path for use decline. I recommend that the Company institute an open depreciation study which will include participation from public stakeholders to determine the best cost recovery processes and rates.

Q. Accelerated depreciation will increase rates. How can we reduce that impact?

A. Given the recent tax law changes, it has become clear that previous rate payers paid too much for deferred Federal Income Taxes. Given that we now know that public policy and law will result in Average Service Lives which are shorter than originally expected, it is clear that previous ratepayers under-paid for asset depreciation. Given that we have both over-payments and under-payments, it would make sense to use the over-payments by previous ratepayers to balance out, as much as possible, the under-payments by those same ratepayers. Thus, the Excess Deferred Federal Income Tax, which might otherwise be returned to ratepayers as lower rates, should be used instead to reduce the depreciation deficiencies which exist now and which will grow in the future.

Q. If cost recovery is accelerated, won't this create windfall profits for the Shareholders?

A. No. Accelerated depreciation does not create a profit for shareholders. It simply converts the shareholders share of asset values into an equivalent share of capital. If anything, one might argue the shareholders would prefer that cost-recovery not be ac-

celerated since doing so reduces the amount of equity which is subject to the allowed return on equity. However, failing to accelerate cost-recovery will result in stranded assets. Thus, the shareholders are better off getting their capital returned to them in a form that allows it to be reinvested in assets that are more likely to deliver acceptable returns over time.

Q. If the amount of equity invested in gas assets is reduced, how do the shareholders get their required return?

A. The Company will either have to find ways to invest the equity in non-gas assets or return the equity to its shareholders through dividends or share buy-backs.

Q. Other than gas assets, what could the Company invest in?

A. As noted before, the company isn't really in the business of selling gas. They make their money by investing in pipes in the ground. While the gas-pipe business is going away, the pipes-in-the-ground business need not. What needs to change is what is in the pipes. The Company should: **Pump heat, not gas.**

It is clear that district geothermal heating schemes have the potential to serve much of the need once served by gas space and water heating. Such systems are quite common in Europe and the Company has recently operated pilots exploring the technology. As has been shown, the key capital investment in such systems is that which is made in the ground loops and distribution pipes.

Like a gas system, a geothermal district heating system is heavily invested in pipes-in-the-ground. It should also be noted that the pipes required for district heating systems will have Average Service Lives which are similar to, if not longer than, the Average Service Lives of the gas pipes that the Company has traditionally invested in but no longer should. Unlike gas pipes, the installation of district heating systems fits

well within existing and anticipated law and policy.

Q. This all sounds complicated. Why can't we wait to deal with these issues later?

A. The longer we allow the accumulation of gas asset to continue, the more expensive it will be for ratepayers to support the gas system in the future. The key problem here is that once a new gas capital investment is made, its costs are essentially fixed over the remaining life of the asset. For instance, if the Company invests in a new service connection with a sixty-year life, it will recover one sixtieth of its up-front cost each year, plus the cost of financing the remaining un-recovered amount. Then, given that the Company uses volumetric rates to recover its costs of delivery, the fixed costs are spread over the number of units of gas sold each year.

Q. Explain why fixed costs are such a problem when sales decline.

A. As long as the number of units of gas sold remains unchanged, then the cost-recovered per unit will stay essentially the same each year until costs are fully recovered. However, if sales decline, then each year the fixed asset costs will be spread over a smaller number of units of gas than during the year before. That means that after a sales decline, the per-unit cost recovery will increase.

For example, if gas consumption were to decline by 80% between now and 2050, then the per-unit cost recovery for an asset installed in 2019 but still in use in 2050 would be 500% greater than it was in 2019.[12] This is because, in 2050, the fixed costs would be spread over just 20% of the number of units sold in 2019. If the decline in sales by 2050 is only 50%, then the per-unit cost would increase by a still substantial 200%.

Q. What is the likely result of the per-unit cost increases that will be driven by reduced gas sales?

[12] Compute the increase by dividing one by one minus the percentage decline. In this case, the formula would be: $\frac{1}{(1-.80)} = \frac{1}{.2} = 5$

A. The result will be a gas **"death-spiral."**

As gas' per-unit costs increase, alternatives to gas will become more competitive and those ratepayers who have the capital or credit needed to invest in alternatives to gas, such as heat pumps, will abandon gas and find shelter with the cheaper alternatives. However, by abandoning gas, they will drive sales even lower and that will increase the per-unit costs to the point where even more ratepayers decide to abandon gas. The cycle then repeats. As the per-unit costs increase, gas sales will decrease even faster, forcing the per-unit costs increases to accelerate. This is referred to as a "death-spiral."

In time, we will find that gas will become known as the "poor-man's fuel" since the remaining gas users will increasingly be those who can't afford to abandon it — no matter how much they might wish to. Low and moderate income ratepayers, who have the most limited financial resources and the greatest need for more affordable energy, will end up carrying the burden of the stranded assets left behind by those who are financially more fortunate.

Q. Does that conclude your direct testimony in this case?

A. Yes.